ORCA vs the Field — SIH26176 Competitive Analysis

Sep 28, 2026 · @nim

Bottom line

ORCA can still win, but its planned headline idea — a deterministic safety gate with an LLM that only explains — is now what every strong team does. The one thing no team does is get a reasoned, boat-specific warning to a fisherman who is already at sea with no mobile signal.

Five moves decide it:

1. **Lead with the last mile at sea**, framed as making ISRO's existing NavIC / DAT-SG channel smarter, not as a new invention. Judges from ISRO know that channel exists.
2. **Win on honesty.** Add plausibility checks, a check that the LLM's words match the engine's verdict, and a data-freshness gate. The flashiest competitor demos have the most visible data errors.
3. **Show brief compliance on screen.** Map the brief's nine agents one-to-one, put its eight example queries as chips, and make query #7 (why productivity fell) the showpiece.
4. **Match the new baseline cheaply:** evidence cards, a live agent panel, a departure-time counterfactual, vessel-class thresholds, Bhashini voice in and out.
5. **Tell a different story.** Two teams open with the same "32,000 deaths / Cyclone Ockhi" script. Open instead with what Ockhi actually exposed: warnings existed but could not reach boats already at sea.

What was analysed

Six pitch videos (transcripts plus slides or screenshots) and nine public GitHub repos for SIH26176, all read on 28 Sep 2026. Some teams appear in both, which allows checking what they claim on stage against what their code says.

Team / repo	Seen via	Stage reached
Slayy Stack (FisherNav)	Video	Slides + UI screenshots
IndusMind	Video	Mock dashboard
Game of Codes	Video	Architecture slides only

Team / repo	Seen via	Stage reached
Unnamed "ORCA v1.0" dashboard	Video	Working demo
Team Kanyakumari	Video	Working demo, simulated data
DeTABIS	Video + repo	Deployed web app
Team Random	Repo	Working app, live weather
Team OrbitX	Repo	Deployed, 258 tests
Shrikant-Dapke	Repo	Working MVP, 223 tests
Tejavelagala	Repo	Working app, Docker stack
Mr1823	Repo	Working app, 176 tests
Team WE# (SagarDrishti)	Repo	Deployed web app
FishingFriend	Repo	Deployed Streamlit app
Tanveer-rajpurohit	Repo	Frontend only, mock data

At least four more SIH26176 repos exist that were not opened. The picture below is representative, not exhaustive.

Team-by-team teardown

The real threats are Game of Codes (clearest reasoning), Team Random (most complete working product) and Kanyakumari and DeTABIS (strongest demos). The rest are useful mainly for single ideas.

Tier 1 — genuine threats

Team Random (repo) — the most complete build seen.

- Ten agents; a documented weighted risk model with deterministic floors that can only raise the score (active severe warning → 92/100, IMD fishermen warning → 70).
- Every number carries source, timestamp, confidence and mode; demo data is stamped as demo.
- A separate phone mode for low-literacy users: big colour verdict, one-tap LISTEN, microphone as the main input. Marathi, Hindi, English.
- Candidate fishing grounds inside protected or defence zones are filtered out before ranking.
- Honest data table: says openly that INCOIS, IMD and MOSDAC have no open JSON API, and uses Open-Meteo as the live source.

- Gap: nothing reaches a boat at sea. Offline PWA and SMS/IVR are only on their roadmap.

Game of Codes (video) — the best-argued pitch.

- "Integration, not invention": they build a reasoning layer over INCOIS, IMD, Bhuvan, MOSDAC and NDMA, not new forecast models.
- Evidence gate outside the LLM (PASS / QUALIFIED / BLOCKED); an official warning forces BLOCK; conflicting sources are shown, never averaged.
- One slide shows query-adaptive planning: for a safety question, Weather, Sea-State, Hazard and Safety are ticked; PFZ and Geospatial are crossed out.
- Departure-time counterfactual ("what if I leave at 7 instead of 5?").
- Gaps: no product shown, no fisherman in the story, pull-only (no proactive alerts), English plus one language.

Team Kanyakumari (video) — the best complete package on video.

- Opens with real Cyclone Ockhi news footage.
- Agent cards each show a one-line output; evidence cards show value, threshold, source tag and contribution to the risk score.
- Separate fisherman and authority portals; one-click demo logins for judges.
- Gaps: all data simulated; thresholds unrealistic (wind "≤5 km/h low risk"); authority view relies on AIS that small boats lack; alerts only in-app.

DeTABIS (video + repo) — the most polished visuals.

- Bhashini ASR for voice input, live voice mode in five languages; answers adapt to vessel class (<8 m, 8–15 m, >15 m) and to role (fisher vs scientist).
- Query chips mirror the brief's eight example questions.
- Gaps: the on-screen source for its PFZ answer is an "illustrative" bathymetry model retrieved in June 2024, while the voiceover claims live INCOIS data. Map shows "API KEY REQUIRED" watermarks. Agent counts disagree (3 / 5 / 6). Its README lists the wrong ministry and puts five non-Indian fishing grounds on the globe.

Tier 2 — strong engineering, weaker pitch

Shrikant-Dapke (repo) — the most rigorous about data honesty.

- Verified that INCOIS ERDDAP's freshest usable SST ends April 2025; enforces a 96-hour freshness gate and reports "unavailable" rather than showing old data as live.
- Verified that INCOIS PFZ has no machine-readable API and MOSDAC needs SSO credentials; both are honest stubs.
- The LLM's verdict must equal the engine's verdict or it is discarded; 223 tests.

Tejavelagala (repo) — the widest feature set.

- Five risk states: SAFE, CAUTION, UNSAFE, BLOCKED, DATA_UNAVAILABLE.
- Last-known-good offline snapshot, SMS/IVR adapter endpoints, human-in-the-loop escalation queue for Coast Guard and INCOIS, Telugu support, a Bhashini-ready adapter.

Mr1823 (repo) — two rules worth copying.

- Alerts are default-deny: an unrecognised alert type forces at least CAUTION.
- A missing reading forces at least CAUTION and is reported, never invented. Tamil support.

Team OrbitX (repo) — deployed, 258 passing tests, Sarvam voice in 11 languages, IMBL and EEZ geofencing, SOS/MAYDAY script generator.

Tier 3 — single ideas worth noting

- **"ORCA v1.0" dashboard** (video): the "collective impact" idea — spread boats across zones so everyone does not rush the best one. But the demo showed SST of 2.6°C in the Bay of Bengal and contradictory GO / CAUTION labels.
- **FisherNav / Slayy Stack** (video): offline-first story and GEBCO bathymetry for reef and grounding risk. But its cyclone data comes from NOAA and OpenWeather, with no IMD.
- **IndusMind** (video): a verification agent that sends failed checks back to the planner, and a "what is normal / changing / unusual" answer structure. But it analyses the Pacific and Atlantic, not Indian fishing.
- **SagarDrishti / Team WE#** (repo): ten languages and IMD port danger signals, but heavy on claims (AIS, IMO FSA, live crawlers) that a judge can puncture.

Comparison matrix

No team has alerts that reach a boat with no mobile signal, and none maps its agents to the brief's nine. Those two columns are ORCA's clearest openings.

Team	Working product	Safety gate outside the LLM	Honest about live vs demo	Proactive alerts	Reaches boat with no signal	Languages / voice	Brief's 9 agents mapped
Team Random	Yes	Yes, with floors	Yes	In-app	No (roadmap)	3, voice in/out	No (10 own agents)
Game of Codes	No	Yes (PASS / QUALIFIED / BLOCKED)	Claimed	No	No	English + 1	No (6 capabilities)

Team	Working product	Safety gate outside the LLM	Honest about live vs demo	Proactive alerts	Reaches boat with no signal	Languages / voice	Brief's 9 agents mapped
Kanyakumari	Yes, simulated	Yes	Labelled	In-app	No	Mobile only	No (6)
DeTABIS	Yes	Partial (cyclone = caution only)	Contradicts itself	Claimed	No	5, Bhashini ASR	No (3–6)
Shrikant-Dapke	Yes	Yes, verdict-match check	Best in field	No	No	3, browser voice	No
Tejavelagala	Yes	Yes, 5 states	Yes	Alert watch	Offline snapshot only	3, browser voice	No
Mr1823	Yes	Yes, default-deny	Yes	No	No	2 (EN, TA)	No
OrbitX	Yes, deployed	Yes	Yes	Notifications	No	11, Sarvam voice	No (11 own)
"ORCA v1.0"	Yes	Claimed	Data errors on screen	Emergency alarm	No	English	No (11 own)
FisherNav	Partial	No	Overclaims	Claimed	Offline cache only	6, TTS	No (4)
IndusMind	Mock	Verification loop	Mock, unlabelled	No	No	English	No (17 own)
ORCA (planned)	—	Yes, regional thresholds + cyclone veto	`threshold_unavailable` state	Yes	**Yes, via NavIC / DAT-SG**	Bhashini, pilot-state languages	Should be

"Claimed" means stated in a pitch or README but not shown working.

The new baseline

These are now table stakes. Having them earns no credit; missing any one costs points, because judges will have seen them in several other teams.

- **The LLM never decides safety.** A deterministic engine produces the verdict; the LLM only words it. Every serious team says this.
- **An official warning overrides everything.** Cyclone or severe IMD warning forces the worst verdict.
- **Evidence with provenance.** Each value shows its source and timestamp, in a "why did ORCA say this?" panel.
- **A visible agent trace.** Which agents ran and what each one found.

- **Map with PFZ, route and geofence layers** (IMBL, protected areas).
- **Voice in and out in at least three Indian languages.**
- **A departure-time counterfactual** and vessel-size awareness are becoming common among the stronger teams.

ORCA's cyclone veto, regional thresholds and `threshold_unavailable` state are good, but on their own they now read as "also has a safety gate". They need to be shown as sharper versions of the baseline, not as the headline.

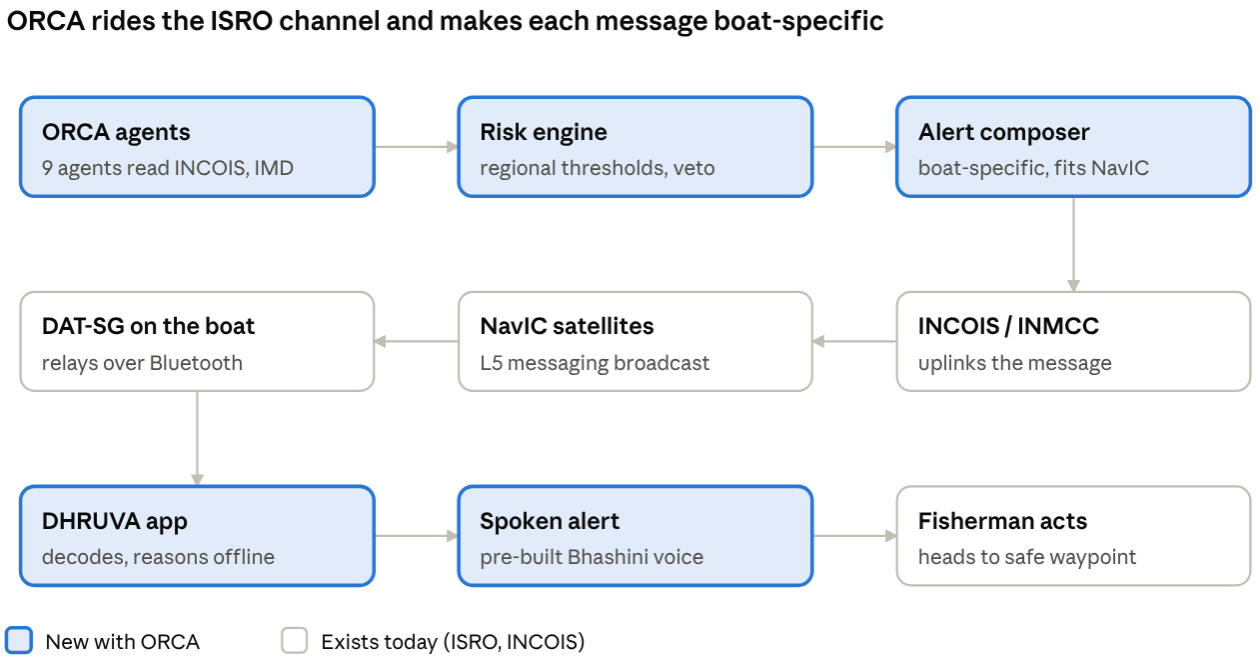
The strategic reframe: ISRO's at-sea channel already exists

ORCA's satellite alert path is not new infrastructure, and ISRO judges will know that. What the existing channel lacks is reasoning: today every boat receives the same broadcast.

What already exists:

- INCOIS uses the NavIC messaging service to broadcast PFZ messages and cyclone and high-wave alerts to fishermen (ISRO at UN ICG, 2022).
- ISRO's DAT-SG, launched January 2024, receives control-centre messages at sea, sends distress alerts with acknowledgement, and relays messages to a phone app over Bluetooth (Tribune).
- ISRO publishes the exact Bluetooth packet format between DAT-SG and the phone; INCOIS and control-room broadcasts start with `$PIRNSF` (ISRO ICD v1.2).
- GEMINI does the same over GAGAN with an INCOIS app in nine languages, launched in 2019 after Ockhi (Scroll).

The pitch line: today a cyclone broadcast says the same thing to every boat. ORCA turns it into "Your boat is inside the warning area. Head south-west to the safe waypoint, 22 km, before 4 PM" — spoken in Malayalam.



ORCA alert path · 3 new layers on top of ISRO's existing channel

ORCA adds three layers, shown highlighted: reasoning on shore, the alert composer, and the app and voice on the boat. The grey boxes are ISRO and INCOIS systems that already run. ORCA cannot transmit on its own, so the demo should replay a simulated DAT-SG Bluetooth stream built to the ISRO packet format.

Two design consequences follow:

- **Messages must be tiny.** NavIC messaging carries very short packets, so the composer sends codes (zone, severity, action, waypoint) and DHRUVA expands them using context cached at the dock. This is what ORCA's 145-bit alert already does.
- **Voice at sea cannot call Bhashini**, because there is no internet. Generate voice templates in each pilot language with Bhashini on shore, ship them in the app, and fill in numbers offline.

Open question to verify before the pitch: the DAT-SG fleet size. News reports from January 2024 cite over 20,000 first-generation DATs in use, while ORCA's notes say 40,000+.

Best solutions: what ORCA should build

Each item below closes a gap no competitor fills, or matches the baseline more convincingly. They are ordered by how much they separate ORCA from the field.

1. Boat-specific alerts at sea (the headline)

- **Alert composer agent.** Takes an official warning (IMD cyclone, INCOIS high-wave) plus each registered boat's last known position, vessel class and home port. Outputs a compact coded message: zone, severity, action, waypoint.
- **DHRUVA decoder.** Parses the ISRO DAT-SG Bluetooth packet format, expands the codes using the map, thresholds and port list cached at the dock, and computes distance and bearing to the safe waypoint offline.
- **Offline reasoning at sea.** With GPS and the cached forecast, DHRUVA can still answer "am I inside the warning area?" and "which way to safety?" with no signal.
- **Voice without internet.** Pre-generate phrase templates in Tamil, Malayalam and Odia with Bhashini; fill in numbers and directions on the phone.
- **Demo:** replay a scripted DAT-SG stream, switch the phone to airplane mode, and have the alert arrive and speak. No other team can show this.

2. The strictest honesty in the field

Combine the best rules from three competitors into one visible "trust layer":

- **Freshness gate** (from Shrikant-Dapke): data older than its validity window becomes "unavailable", never shown as live.
- **Plausibility bounds:** SST in Indian waters roughly 20–32°C, sane ranges for wave and wind. Anything outside becomes "unavailable". This would have caught the "ORCA v1.0" demo's 2.6°C.
- **Default-deny** (from Mr1823): an unrecognised alert or a missing reading forces at least CAUTION.
- **Verdict-match check** (from Shrikant-Dapke): if the LLM's text states a different verdict or number than the engine, discard it and use the template answer.
- **Five states, not three** (from Tejavelagala): SAFE, CAUTION, UNSAFE, BLOCKED (official warning), DATA_UNAVAILABLE. This fits ORCA's existing `threshold_unavailable` idea.

3. Brief compliance you can see

- **Nine-agent map.** One screen or slide naming the brief's nine agents (planning, marine data discovery, weather intelligence, ocean analytics, geospatial reasoning, risk assessment, visualization, reporting, user interaction) with ORCA's implementation of each. No team does this.
- **Eight query chips** matching the brief's example questions, each answering correctly.

- **Query #7 as the showpiece.** Answer "why did fish productivity decline here?" with IndusMind's structure: what is normal, what changed (SST up, chlorophyll down over weeks), what is unusual, and what the fisherman should do. Use a verification step that sends a failed evidence check back to the planner.

4. Match the baseline, sharper

- **Query-adaptive agent panel** (Game of Codes + Kanyakumari): for each question, show which agents ran, which were skipped, and one line from each.
- **Evidence cards:** value, threshold, source, timestamp, and how much it moved the verdict. ORCA's thresholds come with a stated source; most competitors' do not.
- **Departure-time counterfactual:** "what if I leave at 7 instead of 5?" re-runs only the time-dependent agents and shows the verdict change.
- **Vessel class and region** set the thresholds (from DeTABIS), which fits ORCA's regional thresholds.
- **Two views:** a phone view built for low-literacy use (Team Random's big verdict and LISTEN button) and an authority view that can issue the boat-specific alert in step 1.

5. Ideas to consider, not to headline

- **Crowding across fishing zones** ("ORCA v1.0"): only if a real boat-density source exists; DAT-SG position reports might be one, but that needs checking.
- **Grounding and reef risk from GEBCO bathymetry** (FisherNav): cheap to add as a map layer.
- **Fishing-ban calendar and protected-area filter** (Team Random, OrbitX): ranks out zones that are closed or illegal before recommending them.

Data reality: what is actually live

Several competitors independently found that the official Indian sources have no open machine-readable feed. A judge who knows this will test any team claiming "live INCOIS data", so ORCA should say exactly what is live and what is not.

Source	What competitors found	What ORCA should do
INCOIS PFZ advisories	No machine-readable API; text bulletins and WebGIS only (Shrikant-Dapke, Team Random)	Use the official text advisories, check their terms of use, and ask the ISRO sponsor or mentor for a data-sharing route. Label clearly.

Source	What competitors found	What ORCA should do
INCOIS ERDDAP (SST, chlorophyll)	Public, but freshest usable SST grid ends April 2025 (Shrikant-Dapke)	Re-check dataset end dates yourselves; apply a freshness gate. Do not call old data live.
IMD marine warnings	No open JSON API; bulletins only (Team Random, Tejavelagala)	Parse bulletins, and in the demo use a real archived bulletin, labelled with its date.
ISRO MOSDAC	Registration or SSO required (Shrikant-Dapke, Team Random)	Register now so at least one ISRO satellite product is genuinely live by the finale.
Open-Meteo marine	Keyless and live; most working prototypes rely on it	Fine as a fallback, but always labelled "Open-Meteo", never as INCOIS or IMD.
NavIC / DAT-SG messaging	Packet format is public (ISRO ICD v1.2); uplink is ISRO-operated	Build the decoder to spec; simulate the stream in the demo and say so.
Bhashini	Team already has access	Use for voice in and out on shore, and to pre-build offline voice templates.

These findings come from other teams' repos, not from ORCA's own tests. Re-verify each one before the finale; a genuinely live ISRO or INCOIS feed would put ORCA ahead of every repo reviewed.

Pitch and demo playbook

The story should run from Ockhi's real failure to ORCA's airplane-mode alert, with every other feature serving that arc.

The opening story

Ockhi's lesson is not "fishermen die", which two teams already say. It is that warnings existed but could not reach boats already at sea:

- Most fishermen already at sea had no equipment to receive warnings, and Kanyakumari's administration relied on individual phone calls (The Wire).
- About 3,000 fishermen were at sea before the warning went out, with no way to reach boats 200–500 nautical miles offshore (The News Minute).

- An ISRO official said afterwards that the only solution is two-way communication, which led to the satellite devices that exist today (VillageSquare).

Then the turn: "ISRO built the channel. But it sends every boat the same message. ORCA makes it personal."

Demo order (about 5 minutes)

1. **Hook (30 s):** Ockhi, as above. Ends on the one-line problem.
2. **Ask by voice (60 s):** a Tamil question, "Can I go fishing tomorrow at 5?" The agent panel lights up with skipped agents greyed out, then evidence cards and a spoken verdict.
3. **Counterfactual (20 s):** "What if I leave at 7?" The verdict changes, and the card shows why.
4. **Query #7 (45 s):** "Why is the catch down here?" Normal, changed, unusual, and what to do.
5. **Honesty moment (20 s):** a stale feed shows DATA_UNAVAILABLE instead of a number. Say: "ORCA would rather say it doesn't know."
6. **The at-sea moment (60 s):** switch the phone to airplane mode. A simulated DAT-SG packet arrives, and DHRUVA speaks in Malayalam: inside the warning zone, bearing and distance to safety.
7. **Brief compliance (20 s):** one slide with the nine agents and eight queries, each marked done.
8. **Feasibility (30 s):** "Integration, not invention" — ISRO's channel, INCOIS and IMD data, Bhashini, zero-spend stack, pilot in Tamil Nadu, Kerala and Odisha.

Judge questions to rehearse

Likely question	Answer to give
Doesn't GEMINI or the INCOIS app already do this?	They deliver the same broadcast to everyone. ORCA reasons per boat, answers questions on shore, and works offline at sea.
Is your data live?	Say exactly which sources are live and which are labelled demo (see Data reality).
Where do your thresholds come from?	Regional values with cited sources, IMD national fallback, and an explicit "unavailable" state where no source exists.
Can the LLM override a safety call?	No. The engine decides; a text-verdict mismatch is discarded.

Likely question	Answer to give
Can a NavIC message carry all that?	No, so ORCA sends codes and the app expands them from cached context.
Most fishermen lack smartphones.	Tiered access: DAT-SG terminal with the app, or shore-side alerts through the authority view.

Traps to avoid

Every trap below was visible in at least one competitor, and each one would cost ORCA credibility with an ISRO or INCOIS judge.

Trap	Seen in	How ORCA avoids it
Physically impossible numbers on screen (SST 2.6°C)	"ORCA v1.0"	Plausibility bounds; a domain read-through of every number before recording
Voiceover claims live data the screen contradicts	DeTABIS	Say only what the screen shows; source line on every answer
Verdict and text disagree (GO vs CAUTION, 0/100 "CAUTION")	"ORCA v1.0"	Verdict-match check; one source of truth for labels
Broken map tiles or API-key watermarks in the recording	DeTABIS	Dry run on the recording machine; keyless tiles or cached tiles
Thresholds with no source (wind ≤5 km/h "low risk")	Kanyakumari	Cite each threshold or mark it unavailable
AIS vessel tracking for small boats	Kanyakumari, SagarDrishti	Don't claim it; small craft mostly lack AIS
Foreign data for an ISRO brief (NOAA, CMEMS only)	FisherNav, IndusMind	Lead with INCOIS, IMD, MOSDAC, Bhashini
Scope sprawl across five user groups	Kanyakumari, OrbitX	Two views only: fisherman and authority
Same opening statistic as other teams	Kanyakumari, DeTABIS	Use the Ockhi "warning couldn't reach the boat" angle
Agent counts that change between screens	DeTABIS	One number, matched to the brief's nine

Trap	Seen in	How ORCA avoids it
Claiming a local LLM runs on the boat	"ORCA v1.0"	Rule-based reasoning offline; LLM only on shore

Sources

Competitor videos were reviewed from transcripts and screenshots shared by the team; the repos and pages below were opened on 28 Sep 2026.

Competitor repos

- Team Random — orca-sih26176
- DeTABIS — ORCA
- Team OrbitX — Team-OrbitX-SIH-26
- Shrikant-Dapke — ORCA
- Tejavelagala — ORCA-SIH26176
- Mr1823 — OCRA-SIH
- Team WE# — SagarDrishti-AI
- FishingFriend
- Tanveer-rajpurohit — ORCA

ISRO and INCOIS channels

- NavIC and GAGAN system update, ISRO at UN ICG-16 (2022)
- ICD for DAT-SG emergency messaging via NavIC, v1.2 (ISRO)
- ISRO develops second-generation distress alert transmitter (Tribune, Jan 2024)
- A simple handheld device could save the lives of Indian fisherfolk at sea (Scroll, 2019)

Cyclone Ockhi

- Why are some lives more important, ask Ockhi-hit fisherfolk (The Wire)
- Still no word from almost 900 fisherfolk after Cyclone Ockhi (The News Minute)
- Deepsea fishermen need early warning after Cyclone Ockhi (VillageSquare)